

GREENLEAF NURSERY

Architecture, Data Model & Access Control

1. Collection Descriptions and Schema Justification

The GreenLeaf Nursery application uses MongoDB with Mongoose. The schema separates current, changeable inventory data from historical sales records. User access is controlled through roles and permissions.

Plant

Field	Type	Purpose
_id	ObjectId	Unique plant identifier
name	String	Plant name
species	String	Botanical/species name
category	String/ObjectId	Plant category
description	String	Plant description
price	Number	Current selling price
quantity	Number	Current stock quantity
attributes	Object	Flexible plant-specific attributes
supplier	String/ObjectId	Supplier reference
createdAt / updatedAt	Date	Audit timestamps

Justification: Plant is an independent collection because its price, stock, description and availability may change. A flexible attributes object accommodates characteristics that differ between plant types without creating separate collections for every plant category.

Sale

Field	Type	Purpose
_id	ObjectId	Unique sale identifier
customerName	String	Customer name
customerContact	String	Customer contact
items	Array/ObjectId	Purchased sale items

totalAmount	Number	Final transaction amount
paymentMethod	String	Payment method
createdBy	ObjectId	User who processed the sale
createdAt	Date	Transaction timestamp

Justification: A completed sale is a historical record. It should not change merely because the current plant price or description changes.

SaleItem

Field	Type	Purpose
_id	ObjectId	Unique line-item identifier
saleId	ObjectId	Parent sale reference
plantId	ObjectId	Plant reference
plantName	String	Name captured at sale time
unitPrice	Number	Price captured at sale time
quantity	Number	Quantity purchased
subtotal	Number	Unit price multiplied by quantity

Justification: SaleItem combines a reference to the plant with a historical snapshot. If a plant costs Rs. 1,500 when sold and later costs Rs. 1,800, the original SaleItem continues to record Rs. 1,500.

User

Field	Type	Purpose
_id	ObjectId	Unique user identifier
name	String	User name
email	String	Login email
passwordHash	String	Hashed password

roleId	ObjectId	Reference to Role
isActive	Boolean	Account status
createdAt	Date	Account creation timestamp

Justification: Users reference roles instead of duplicating permission definitions. This centralizes authorization rules and makes role changes easier to manage.

Role

Field	Type	Purpose
_id	ObjectId	Unique role identifier
name	String	Admin, Manager or Staff
permissions	Array	Permission identifiers
description	String	Role description

Justification: Roles provide centralized RBAC. Permissions such as plant.create or sale.read can be assigned to a role without changing every user document.

2. Embedded vs Referenced Schema Trade-off

Approach	Advantages	Disadvantages	GreenLeaf Use
Embedded	Simple reads; related data can be retrieved together	Duplication; document growth; harder independent updates	Small, tightly coupled snapshots
Referenced	Independent updates; reusable entities; better separation	More queries/lookups; joins through application logic or aggregation	Plant, User, Role, Sale relationships
Hybrid	Balances normalization and historical snapshots	Requires clear ownership rules	Recommended overall design

Recommended design: reference entities that change independently, while embedding/capturing historical values needed to reproduce a completed transaction.

3. RBAC Matrix

Resource / Action	Admin	Manager	Staff
View plants	Yes	Yes	Yes
Add plants	Yes	Yes	No
Update plants	Yes	Yes	Limited
Delete plants	Yes	Yes	No
View inventory	Yes	Yes	Yes
Update stock quantity	Yes	Yes	Limited
View sales	Yes	Yes	Assigned/operational
Create sale	Yes	Yes	Yes
Update completed sale	Yes	Limited	No
Cancel/delete sale	Yes	Limited	No
View users	Yes	No	No
Create users	Yes	No	No
Update users	Yes	No	No
Deactivate users	Yes	No	No
Manage roles	Yes	No	No
View reports	Yes	Yes	Limited
Export reports	Yes	Yes	No
System configuration	Yes	No	No

Authorization must be enforced by the Express backend, not only by hiding frontend buttons. A protected request should pass JWT authentication and then a permission check before the controller/service layer is allowed to execute.

4. Recommended Permission Model

plant.read, plant.create, plant.update, plant.delete

sale.read, sale.create, sale.update, sale.cancel

user.read, user.create, user.update, user.delete

report.read, report.export

5. Data Integrity Rule

Historical sales must not be recalculated from the current Plant document. SaleItem stores the unit price and relevant identifying information at the time of purchase. Current inventory data remains in Plant, while completed transaction data remains stable.